Chapter 7. Complete Metric Spaces and Function Spaces

Completeness and completion, compactness in metric spaces, compact convergence, Ascoli's theorem

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7.1 Complete metric spaces

Motivation. Completeness, every Cauchy sequence converges, is the property that makes limits exist without being given in advance. It is not a topological property (it depends on the metric), but function spaces with the uniform metric are complete, and every metric space can be completed.

A sequence (x_n) in (X, d) is **Cauchy** if for every $\epsilon > 0$ there is N with $d(x_n, x_m) < \epsilon$ for $n, m \geq N$; (X, d) is **complete** if every Cauchy sequence converges. A closed subset of a complete space is complete, and a complete subspace of a metric space is closed. (X, d) is complete iff it is complete in the standard bounded metric $\bar{d} = \min(d, 1)$, since the two have the same Cauchy sequences.

Lemma 7.1 (43.1)

A metric space is complete if every Cauchy sequence has a convergent subsequence.

Proof. Let (x_n) be Cauchy with $x_{n_i} \to x$. Given ϵ, choose N with $d(x_n, x_m) < \epsilon/2$ for $n, m \geq N$, and i with $n_i \geq N$ and $d(x_{n_i}, x) < \epsilon/2$. Then $d(x_n, x) < \epsilon$ for $n \geq N$. ■

Theorem 7.2 (43.2)

$\mathbb{R}^k$ is complete in the euclidean metric d and in the square metric ρ.

Proof. Since $\rho \leq d \leq \sqrt{k}\rho$, the two metrics have the same Cauchy sequences; use ρ. A Cauchy sequence is bounded, so it lies in a cube $[-M, M]^k$, which is compact (Theorem 3.26), hence sequentially compact (Theorem 3.33). Apply Lemma 7.1. ■

Lemma 7.3 (43.3)

In $X = \prod X_\alpha$ (product topology), $x_n \to x$ iff $\pi_\alpha(x_n) \to \pi_\alpha(x)$ for each α.

Proof. ($\Rightarrow$) Each π_α is continuous.

($\Leftarrow$) A basis element $\prod U_\alpha$ containing x restricts only $\alpha_1, \ldots, \alpha_k$; choose N_i with $\pi_{\alpha_i}(x_n) \in U_{\alpha_i}$ for $n \geq N_i$, and let $N = \max N_i$. ■

Theorem 7.4 (43.4)

There is a metric on $\mathbb{R}^\omega$ (product topology) in which it is complete.

Proof. Take $D(x, y) = \sup_i \bar{d}(x_i, y_i)/i$, which induces the product topology (Theorem 2.36). If (x^n) is D-Cauchy, then $\bar{d}(x_i^n, x_i^m) \leq iD(x^n, x^m)$, so each coordinate sequence is Cauchy in $\mathbb{R}$ and converges. By Lemma 7.3, (x^n) converges in $\mathbb{R}^\omega$. ■

For a metric space (Y, d) and a set J, the **uniform metric** on Y^J is $\bar{\rho}(x, y) = \sup_\alpha \bar{d}(x_\alpha, y_\alpha)$.

Theorem 7.5 (43.5)

If Y is complete in d, then Y^J is complete in the uniform metric $\bar{\rho}$.

Proof. Y is complete in $\bar{d}$. If (f_n) is $\bar{\rho}$-Cauchy, then $\bar{d}(f_n(\alpha), f_m(\alpha)) \leq \bar{\rho}(f_n, f_m)$, so each $f_n(\alpha)$ converges, say to $f(\alpha)$. Given ϵ, choose N with $\bar{\rho}(f_n, f_m) < \epsilon/2$ for $n, m \geq N$. For fixed $n \geq N$ and α, letting $m \to \infty$ gives $\bar{d}(f_n(\alpha), f(\alpha)) \leq \epsilon/2$; hence $\bar{\rho}(f_n, f) \leq \epsilon/2 < \epsilon$. ■

Let $\mathcal{C}(X,Y)$ be the continuous and $\mathcal{B}(X,Y)$ the bounded functions $X \to Y$. On $\mathcal{B}(X,Y)$ the **sup metric** is $\rho(f,g) = \sup_x d(f(x),g(x))$; there $\bar{\rho} = \min(\rho, 1)$.

Theorem 7.6 (43.6)

Let X be a space and (Y,d) a metric space. $\mathcal{C}(X,Y)$ and $\mathcal{B}(X,Y)$ are closed in Y^X under the uniform metric. Hence if Y is complete, they are complete in $\bar{\rho}$, and $\mathcal{B}(X,Y)$ is complete in the sup metric ρ.

Proof. If $f_n \to f$ in $\bar{\rho}$ with f_n continuous, then $f_n \to f$ uniformly (for $\epsilon < 1$, $\bar{\rho} < \epsilon$ means $d < \epsilon$ everywhere), so f is continuous (Theorem 2.42). If the f_n are bounded, choose N with $\bar{\rho}(f_N, f) < \frac{1}{2}$; then $d(f_N(x), f(x)) < \frac{1}{2}$ for all x, so $\operatorname{diam} f(X) \le \operatorname{diam} f_N(X) + 1$. Completeness follows from Theorem 7.5, and for ρ because ρ and $\bar{\rho} = \min(\rho, 1)$ have the same Cauchy and convergent sequences. ■

Theorem 7.7 (43.7)

Every metric space (X,d) has an isometric imbedding into a complete metric space.

Proof. $\mathcal{B}(X,\mathbb{R})$ is complete in the sup metric (Theorem 7.6). Fix x_0 and for $a \in X$ define $\phi_a(x) = d(x,a) - d(x,x_0)$; it is bounded by $d(a,x_0)$. Now $|\phi_a(x) - \phi_b(x)| = |d(x,a) - d(x,b)| \le d(a,b)$, with equality at $x = a$; so $\rho(\phi_a, \phi_b) = d(a,b)$ and $a \mapsto \phi_a$ is an isometric imbedding. ■

The closure of the image is a **completion** of X: a complete metric space containing an isometric copy of X as a dense subset. It is unique up to an isometry fixing X. Completeness is not a topological property: $\mathbb{R}$ is complete but the homeomorphic $(-1,1)$ is not.

7.2 A space-filling curve

Motivation. Completeness of the space of continuous functions can be used to construct surprising objects. Here a uniform limit of polygonal paths gives a continuous surjection of $[0,1]$ onto the square, showing that dimension is not preserved by continuous maps.

Theorem 7.8 (44.1)

Let $I = [0,1]$. There is a continuous map $f : I \to I^2$ whose image is all of I^2.

Proof.

Step 1. call a path in a square $[a, a+s] \times [b, b+s]$ *triangular* if it runs linearly from the lower left corner to the centre of the top edge and then to the lower right corner. Given a triangular path in a square, divide the square into four subsquares of side $s/2$ and replace the path by the concatenation of four triangular paths, one in each subsquare, traversed in the order lower left, upper left, upper right, lower right (with orientations chosen so that the path still starts and ends at the original endpoints). Applied to a path made of many triangular paths, the operation is performed on each one.

Step 2. let f_0 be the triangular path in I^2 and f_n the result of applying the operation n times. Then I is divided into 4^n intervals of length 4^{-n}, and on each such interval f_n is a triangular path in a subsquare of side 2^{-n}; the 4^n subsquares are exactly the squares of the grid of side 2^{-n}. On the same interval f_{n+1} stays in the same subsquare. Hence $d(f_n(t), f_{n+1}(t)) \le \sqrt{2}\,2^{-n}$ for all t, and (f_n) is Cauchy in the sup metric on $\mathcal{C}(I, I^2)$, which is complete (Theorem 7.6, I^2 being closed in the complete $\mathbb{R}^2$). Let f be the limit; it is continuous, and $d(f(t), f_n(t)) \le \sum_{k \ge n} \sqrt{2}\,2^{-k} = 2\sqrt{2}\,2^{-n}$.

Step 3. given $x \in I^2$, x lies in some grid square of side 2^{-n}, which f_n visits, so there is t_n with $d(f_n(t_n), x) \leq \sqrt{2}\,2^{-n}$. Then $d(f(t_n), x) \leq 3\sqrt{2}\,2^{-n} \to 0$, so $x \in \overline{f(I)} = f(I)$, the image being compact and hence closed. ■

7.3 Compactness in metric spaces

Motivation. In metric spaces compactness splits into two simpler properties: completeness and total boundedness. This characterization is what we will use to find compact subsets of function spaces.

(X, d) is **totally bounded** if for each $\epsilon > 0$ it is covered by finitely many ϵ-balls.

Theorem 7.9 (45.1)

A metric space is compact iff it is complete and totally bounded.

Proof. ($\Rightarrow$) A Cauchy sequence has a convergent subsequence (Theorem 3.33), hence converges (Lemma 7.1); and finitely many of the ϵ-balls cover.

($\Leftarrow$) We show sequential compactness. Let (x_n) be given. Cover X by finitely many balls of radius 1; one, B_1, contains x_n for n in an infinite set J_1. Given an infinite J_{k-1}, cover X by finitely many balls of radius $1/k$; one, B_k, contains x_n for infinitely many $n \in J_{k-1}$; call them J_k. Choose $n_1 < n_2 < \cdots$ with $n_k \in J_k$. For $i, j \geq k$, $x_{n_i}, x_{n_j} \in B_k$, so $d(x_{n_i}, x_{n_j}) < 2/k$. The subsequence is Cauchy and converges. ■

A family $\mathcal{F} \subset \mathcal{C}(X, Y)$ is **equicontinuous at** x_0 if for each $\epsilon > 0$ there is a neighbourhood U of x_0 with $d(f(x), f(x_0)) < \epsilon$ for all $x \in U$ and all $f \in \mathcal{F}$; it is **equicontinuous** if this holds at every point.

Lemma 7.10 (45.2)

Let X be a space and (Y, d) a metric space. If $\mathcal{F} \subset \mathcal{C}(X, Y)$ is totally bounded under the uniform metric, then $\mathcal{F}$ is equicontinuous under d.

Proof. Let $0 < \epsilon < 1$ and $x_0 \in X$. Cover $\mathcal{F}$ by $\bar{\rho}$-balls $B(f_1, \epsilon/3), \ldots, B(f_n, \epsilon/3)$, and choose a neighbourhood U of x_0 with $d(f_i(x), f_i(x_0)) < \epsilon/3$ for all i and $x \in U$. For $f \in \mathcal{F}$ pick i with $\bar{\rho}(f, f_i) < \epsilon/3$; since $\epsilon/3 < 1$, $d(f(x), f_i(x)) < \epsilon/3$ for all x. For $x \in U$ the triangle inequality gives $d(f(x), f(x_0)) < \epsilon$. ■

Lemma 7.11 (45.3)

Let X and Y be compact, (Y, d) metric. If $\mathcal{F} \subset \mathcal{C}(X, Y)$ is equicontinuous under d, then $\mathcal{F}$ is totally bounded under the sup metric and the uniform metric.

Proof. Since $\bar{\rho} \leq \rho$, it suffices to treat ρ. Let $\epsilon > 0$. For each $a \in X$ choose a neighbourhood U_a with $d(f(x), f(a)) < \epsilon/3$ for $x \in U_a$, $f \in \mathcal{F}$; finitely many $U_{a_1}, \ldots, U_{a_k}$ cover X. Cover Y by finitely many open $V_1, \ldots, V_m$ of diameter less than $\epsilon/3$. For each function $\alpha : \{1, \ldots, k\} \to \{1, \ldots, m\}$ for which some $f \in \mathcal{F}$ has $f(a_i) \in V_{\alpha(i)}$ for all i, choose one such f_α. Given $f \in \mathcal{F}$, take α with $f(a_i) \in V_{\alpha(i)}$ for all i. For $x \in U_{a_i}$,

$$d(f(x), f_\alpha(x)) \leq d(f(x), f(a_i)) + d(f(a_i), f_\alpha(a_i)) + d(f_\alpha(a_i), f_\alpha(x)) < \epsilon.$$

The continuous function $d(f(x), f_\alpha(x))$ attains its maximum on the compact X, so $\rho(f, f_\alpha) < \epsilon$. The finitely many balls $B_\rho(f_\alpha, \epsilon)$ cover $\mathcal{F}$. ■

$\mathcal{F} \subset \mathcal{C}(X, \mathbb{R}^n)$ is **pointwise bounded** if $\mathcal{F}_a = \{f(a) : f \in \mathcal{F}\}$ is bounded for each a.

Theorem 7.12 (45.4, Ascoli's theorem, classical version)

Let X be compact and give $\mathcal{C}(X, \mathbb{R}^n)$ the uniform topology (d the square or the euclidean metric). A subspace $\mathcal{F}$ has compact closure iff $\mathcal{F}$ is equicontinuous and pointwise bounded under d.

Proof. Let $\mathcal{G} = \overline{\mathcal{F}}$; on $\mathcal{C}(X, \mathbb{R}^n)$ the sup metric ρ induces the uniform topology.

($\Rightarrow$) $\mathcal{G}$ is compact, hence totally bounded under ρ, and so is $\mathcal{F}$; by Lemma 7.10, $\mathcal{F}$ is equicontinuous. Evaluation $f \mapsto f(a)$ is continuous, so $\mathcal{G}_a$ is compact, hence bounded, and $\mathcal{F}_a \subset \mathcal{G}_a$.

($\Leftarrow$)

Step 1. $\mathcal{G}$ is equicontinuous and pointwise bounded. Given a and ϵ, choose U with $d(f(x), f(a)) < \epsilon/3$ for $x \in U$, $f \in \mathcal{F}$; for $g \in \mathcal{G}$ choose $f \in \mathcal{F}$ with $\rho(f, g) < \epsilon/3$; then $d(g(x), g(a)) < \epsilon$ on U. Also each $g(a)$ is within 1 of $\mathcal{F}_a$, so $\mathcal{G}_a$ is bounded.

Step 2. the union of the images $g(X)$, $g \in \mathcal{G}$, lies in a compact $Y \subset \mathbb{R}^n$. Choose neighbourhoods U_a with $d(g(x), g(a)) < 1$ for $x \in U_a$, $g \in \mathcal{G}$, and cover X by $U_{a_1}, \ldots, U_{a_k}$. Every $g(x)$ is within 1 of the bounded set $\mathcal{G}_{a_1} \cup \cdots \cup \mathcal{G}_{a_k}$, so all images lie in a closed bounded cube Y, compact by Theorem 3.26.

Step 3. $\mathcal{G} \subset \mathcal{C}(X, Y)$ is closed in the complete space $\mathcal{C}(X, \mathbb{R}^n)$ (Theorems 7.2, 7.6), hence complete; it is totally bounded by Lemma 7.11; hence compact by Theorem 7.9. ■

Corollary 7.13 (45.5)

Let X be compact. A subspace $\mathcal{F}$ of $\mathcal{C}(X, \mathbb{R}^n)$ (uniform topology) is compact iff it is closed, bounded under the sup metric ρ, and equicontinuous under d.

Proof. A compact $\mathcal{F}$ is closed and bounded in the metric space, and equicontinuous by Theorem 7.12. Conversely, if $\mathcal{F}$ is closed it equals its closure; ρ-boundedness implies pointwise boundedness; apply Theorem 7.12. ■

Example 7.1

Let $f_n : [0, 1] \to \mathbb{R}$ be differentiable with $|f_n(0)| \leq 1$ and $|f_n'| \leq 1$. By the mean value theorem $|f_n(x) - f_n(y)| \leq |x - y|$ (equicontinuity) and $|f_n| \leq 2$ (pointwise boundedness), so some subsequence converges uniformly. The family $\{x^n\}$ on $[0, 1]$ is bounded but not equicontinuous at 1, and indeed has no uniformly convergent subsequence.

7.4 Pointwise and compact convergence

Motivation. Different modes of convergence of functions correspond to different topologies on function spaces. Uniform convergence is often too strong and pointwise too weak; convergence on compact sets is the one that preserves continuity and fits analysis.

For a set X and space Y, the **topology of pointwise convergence** on Y^X has subbasis $S(x, U) = \{f : f(x) \in U\}$ ($x \in X$, U open in Y); it is the product topology.

Theorem 7.14 (46.1)

$f_n \to f$ in the topology of pointwise convergence iff $f_n(x) \to f(x)$ for each $x \in X$.

Proof. This is Lemma 7.3 for the product topology. ∎

Let X be a space and (Y,d) a metric space. For $f \in Y^X$, compact $C \subset X$ and $\epsilon > 0$ let $B_C(f,\epsilon) = \{g : \sup_{x\in C} d(f(x),g(x)) < \epsilon\}$. These sets form a basis: if $g \in B_C(f,\epsilon)$ and $\delta = \epsilon - \sup_C d(f,g)$, then $B_C(g,\delta) \subset B_C(f,\epsilon)$, and if g lies in $B_{C_1}(f_1,\epsilon_1) \cap B_{C_2}(f_2,\epsilon_2)$ then, with the corresponding δ_1, δ_2, $B_{C_1\cup C_2}(g, \min \delta_i)$ lies in both. The resulting topology is the **topology of compact convergence**.

Theorem 7.15 (46.2)

$f_n \to f$ in the topology of compact convergence iff for each compact $C \subset X$, $f_n|C \to f|C$ uniformly.

Proof. Every basis element containing f contains some $B_C(f,\epsilon)$, so convergence means: for every C and ϵ, eventually $\sup_C d(f_n,f) < \epsilon$. That is uniform convergence on each C. ∎

X is **compactly generated** if $A \subset X$ is open whenever $A \cap C$ is open in C for every compact C (equivalently for closed sets).

Lemma 7.16 (46.3)

If X is locally compact, or first countable, then X is compactly generated.

Proof. Suppose $A \cap C$ is open in C for every compact C, and $x \in A$. If X is locally compact, take a compact C containing a neighbourhood U of x; then $A \cap U$ is open in U, hence in X, and $x \in A \cap U \subset A$. If X is first countable and A is not open, some $x \in A$ has no neighbourhood inside A; with a nested countable basis $B_1 \supset B_2 \supset \cdots$ at x, pick $x_n \in B_n - A$. Then $x_n \to x$, and $C = \{x\} \cup \{x_n\}$ is compact. The set $A \cap C$ is open in C and contains x, so it contains x_n for large n, a contradiction. ∎

Lemma 7.17 (46.4)

If X is compactly generated, then $f : X \to Y$ is continuous whenever $f|C$ is continuous for every compact $C \subset X$.

Proof. For open $V \subset Y$, $f^{-1}(V) \cap C = (f|C)^{-1}(V)$ is open in C for every C, so $f^{-1}(V)$ is open. ∎

Theorem 7.18 (46.5)

Let X be compactly generated and (Y,d) metric. Then $\mathcal{C}(X,Y)$ is closed in Y^X in the topology of compact convergence.

Proof. Let f be in the closure of $\mathcal{C}(X,Y)$ and C compact. For each n choose $f_n \in B_C(f,1/n) \cap \mathcal{C}(X,Y)$. Then $f_n|C \to f|C$ uniformly, so $f|C$ is continuous (Theorem 2.42). By Lemma 7.17, f is continuous. ∎

Corollary 7.19 (46.6)

If X is compactly generated, (Y,d) metric, and continuous f_n converge to f in the topology of compact convergence, then f is continuous.

Proof. The limit of a sequence in a set lies in its closure; apply Theorem 7.18. ∎

Theorem 7.20 (46.7)

For Y^X with (Y,d) metric: (uniform) $\supset$ (compact convergence) $\supset$ (pointwise convergence). If X is compact the first two coincide; if X is discrete the last two coincide.

Proof. If $g \in B_C(f,\epsilon)$ and $\delta = \epsilon - \sup_C d(f,g)$, the uniform ball $B_{\bar\rho}(g, \min(\delta,1))$ lies in $B_C(f,\epsilon)$, because $\bar\rho(g,h) < \min(\delta,1)$ forces $d(g(x),h(x)) < \delta$ for all x. If $g \in S(x,U)$, choose ϵ with $B(g(x),\epsilon) \subset U$; then $B_{\{x\}}(g,\epsilon) \subset S(x,U)$. If X is compact, for $\epsilon \le 1$ the uniform ball $B_{\bar\rho}(f,\epsilon)$ is exactly $B_X(f,\epsilon)$. If X is discrete, compact sets are finite and $B_C(f,\epsilon) = \bigcap_{x\in C} S(x, B(f(x),\epsilon))$. ■

For spaces X, Y, the **compact-open topology** on $\mathcal{C}(X,Y)$ has subbasis $S(C,U) = \{f : f(C) \subset U\}$ with C compact and U open.

Theorem 7.21 (46.8)

If (Y,d) is metric, the compact-open topology on $\mathcal{C}(X,Y)$ equals the topology of compact convergence.

Proof. Let $f \in S(C,U)$. The continuous function $y \mapsto d(y, Y-U)$ is positive on the compact $f(C)$ (if $U \neq Y$), so has a positive minimum ϵ; then $B_C(f,\epsilon) \subset S(C,U)$. Conversely consider a basic set $B_C(f,\epsilon)$ with f continuous (it suffices to treat those centred at f). For each $x \in C$ choose a neighbourhood V_x with $f(V_x) \subset B(f(x),\epsilon/4)$, so $f(\overline{V_x}) \subset B(f(x),\epsilon/3)$. Cover C by $V_{x_1},\dots,V_{x_n}$ and put $C_i = \overline{V_{x_i}} \cap C$ (compact), $U_i = B(f(x_i),\epsilon/3)$. Then $f \in W = \bigcap S(C_i,U_i)$. If $g \in W$ and $x \in C_i$, both $g(x)$ and $f(x)$ lie in U_i, so $d(f(x),g(x)) < 2\epsilon/3$. Hence $\sup_C d(f,g) \le 2\epsilon/3 < \epsilon$ and $W \subset B_C(f,\epsilon)$. ■

Corollary 7.22 (46.9)

For metric Y, the topology of compact convergence on $\mathcal{C}(X,Y)$ does not depend on the metric of Y; hence if X is compact, neither does the uniform topology on $\mathcal{C}(X,Y)$.

Proof. The compact-open topology is defined without a metric; use Theorems 7.21 and 7.20. ■

Theorem 7.23 (46.10)

If X is locally compact Hausdorff and $\mathcal{C}(X,Y)$ has the compact-open topology, the **evaluation map** $e : X \times \mathcal{C}(X,Y) \to Y$, $e(x,f) = f(x)$, is continuous.

Proof. Let $f(x) \in V$, V open. By Theorem 3.35 there is a neighbourhood U of x with $\overline{U}$ compact and $\overline{U} \subset f^{-1}(V)$. Then $U \times S(\overline{U}, V)$ is a neighbourhood of (x,f) that e maps into V. ■

Theorem 7.24 (46.11)

Give $\mathcal{C}(X,Y)$ the compact-open topology. If $f : X \times Z \to Y$ is continuous, so is the induced map $F : Z \to \mathcal{C}(X,Y)$, $F(z)(x) = f(x,z)$. The converse holds if X is locally compact Hausdorff.

Proof. Each $F(z)$ is continuous, as a composite $x \mapsto (x,z) \mapsto f(x,z)$. Let $F(z_0) \in S(C,U)$; then $C \times \{z_0\} \subset f^{-1}(U)$, and since C is compact the tube lemma (Lemma 3.21) gives a neighbourhood W of z_0 with $C \times W \subset f^{-1}(U)$; thus $F(W) \subset S(C,U)$. Conversely, $f = e \circ (\mathrm{id}_X \times F)$ is continuous by Theorem 7.23. ■

Thus, for locally compact Hausdorff X, a homotopy $X \times I \to Y$ is the same as a path in $\mathcal{C}(X,Y)$.

7.5 Ascoli's theorem

Motivation. Which families of continuous functions have uniformly convergent subsequences? Ascoli's theorem answers: those that are equicontinuous and pointwise bounded. It is the function-space version of Bolzano–Weierstrass and is used to prove existence theorems for differential equations.

Theorem 7.25 (47.1, Ascoli's theorem)

Let X be a space, (Y,d) a metric space, and $\mathcal{C}(X,Y)$ with the topology of compact convergence; let $\mathcal{F}\subset\mathcal{C}(X,Y)$.
(a) If $\mathcal{F}$ is equicontinuous under d and each $\mathcal{F}_a=\{f(a):f\in\mathcal{F}\}$ has compact closure, then $\mathcal{F}$ lies in a compact subspace of $\mathcal{C}(X,Y)$.
(b) The converse holds if X is locally compact Hausdorff.

Proof. (a) Let $\mathcal{G}$ be the closure of $\mathcal{F}$ in Y^X with the product topology.

Step 1. $\mathcal{G}$ is equicontinuous and $\mathcal{G}_a\subset\overline{\mathcal{F}_a}$. Given a and ϵ, take U with $d(f(x),f(a))<\epsilon/3$ for $x\in U$, $f\in\mathcal{F}$. For $g\in\mathcal{G}$ and $x\in U$, the product neighbourhood $\{h:h(x)\in B(g(x),\epsilon/3),\ h(a)\in B(g(a),\epsilon/3)\}$ of g contains some $f\in\mathcal{F}$, whence $d(g(x),g(a))<\epsilon$. In particular $\mathcal{G}\subset\mathcal{C}(X,Y)$. Since π_a is continuous, $\pi_a(\mathcal{G})\subset\overline{\pi_a(\mathcal{F})}$.

Step 2. $\mathcal{G}$ is compact in the product topology, being closed in $\prod_a\overline{\mathcal{F}_a}$, which is compact by Tychonoff (Theorem 5.3).

Step 3. on $\mathcal{G}$ the product topology and the topology of compact convergence agree. One is coarser (Theorem 7.20); for the other, let $g\in\mathcal{G}$, C compact, $\epsilon>0$. For $x\in C$ choose U_x with $d(h(y),h(x))<\epsilon/3$ for $y\in U_x$, $h\in\mathcal{G}$; cover C by $U_{x_1},\dots,U_{x_n}$, and let $W=\{h\in\mathcal{G}:d(h(x_i),g(x_i))<\epsilon/3\ \forall i\}$, open in the product topology. For $h\in W$ and $y\in C\cap U_{x_i}$, $d(h(y),g(y))<\epsilon$ by three applications of the triangle inequality; the maximum over compact C is $<\epsilon$, so $W\subset B_C(g,\epsilon)$. Therefore $\mathcal{G}$ is a compact subspace of $\mathcal{C}(X,Y)$ containing $\mathcal{F}$.

(b) Let $\mathcal{H}\supset\mathcal{F}$ be compact in $\mathcal{C}(X,Y)$. Evaluation at a is continuous (the pointwise topology is coarser), so $\mathcal{H}_a$ is compact and $\overline{\mathcal{F}_a}\subset\mathcal{H}_a$ is compact. For equicontinuity at a, choose a compact C containing a neighbourhood of a. Restriction $r:\mathcal{C}(X,Y)\to\mathcal{C}(C,Y)$ is continuous when $\mathcal{C}(C,Y)$ has the uniform topology, since $r^{-1}(B_C(r(f),\epsilon))\supset B_C(f,\epsilon)$ and for compact C these balls give the uniform topology. So $r(\mathcal{H})$ is compact, hence totally bounded under $\bar\rho$, hence equicontinuous on C (Lemma 7.10). As C contains a neighbourhood of a, $\mathcal{H}$, and therefore $\mathcal{F}$, is equicontinuous at a. ∎

For $Y=\mathbb{R}^n$, the condition on $\mathcal{F}_a$ is pointwise boundedness, and for compact X the theorem reduces to Theorem 7.12.

7.6 Selected exercises

Motivation. These exercises from Munkres are three standard uses of completeness and compactness in metric spaces: extending uniformly continuous maps to the closure, the contraction principle, and the sequential form of Ascoli's theorem (Arzelà).

Exercise 7.1 (§43, Ex. 2; extension of uniformly continuous maps)

Let (X,d_X) and (Y,d_Y) be metric spaces with Y complete, $A\subset X$, and $f:A\to Y$ uniformly continuous. Show that f extends uniquely to a continuous $g:\overline{A}\to Y$, and that g is uniformly continuous.

Solution.

Step 1. (Images of convergent sequences are Cauchy.) Let $a_n \in A$, $a_n \to x$. Given $\epsilon > 0$, choose $\delta > 0$ with $d_Y(f(a), f(a')) < \epsilon$ whenever $a, a' \in A$, $d_X(a, a') < \delta$. Since (a_n) converges it is Cauchy, so $d_X(a_n, a_m) < \delta$ and hence $d_Y(f(a_n), f(a_m)) < \epsilon$ for n, m large. As Y is complete, $f(a_n)$ converges.

Step 2. (Definition of g.) For $x \in \overline{A}$ there are $a_n \in A$ with $a_n \to x$ (Lemma 2.38); put $g(x) = \lim f(a_n)$. This does not depend on the sequence: if also $a'_n \to x$, the sequence $a_1, a'_1, a_2, a'_2, \ldots$ converges to x, so by Step 1 its image converges, and the subsequences $f(a_n)$, $f(a'_n)$ have the same limit. Using constant sequences, $g = f$ on A.

Step 3. (g is uniformly continuous.) Given $\epsilon > 0$, choose δ as in Step 1 for $\epsilon/2$. Let $x, y \in \overline{A}$ with $d_X(x, y) < \delta$, and $a_n \to x$, $b_n \to y$ in A. Since $d_X(a_n, b_n) \to d_X(x, y) < \delta$, for large n we have $d_X(a_n, b_n) < \delta$ and $d_Y(f(a_n), f(b_n)) < \epsilon/2$. Letting $n \to \infty$ (the metric is continuous), $d_Y(g(x), g(y)) \le \epsilon/2 < \epsilon$.

Step 4. (Uniqueness.) Two continuous extensions agree on A, hence on $\overline{A}$ by Lemma 5.6. ■

Exercise 7.2 (§43, Ex. 5; the contraction principle)

Let (X, d) be a nonempty complete metric space and $f : X \to X$ a **contraction**: there is $\alpha < 1$ with $d(f(x), f(y)) \le \alpha\, d(x, y)$ for all x, y. Show that f has a unique fixed point.

Solution.

Step 1. Fix $x_0 \in X$ and let $x_n = f^n(x_0)$. By induction $d(x_{k+1}, x_k) \le \alpha^k d(x_1, x_0)$. For $m > n$,

$$d(x_m, x_n) \le \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \le \sum_{k=n}^{\infty} \alpha^k d(x_1, x_0) = \frac{\alpha^n}{1-\alpha}\, d(x_1, x_0),$$

which tends to 0 as $n \to \infty$. So (x_n) is Cauchy and converges to some $x \in X$.

Step 2. f is continuous (given ϵ, take $\delta = \epsilon$), so $f(x) = \lim f(x_n) = \lim x_{n+1} = x$.

Step 3. If also $f(y) = y$, then $d(x, y) = d(f(x), f(y)) \le \alpha\, d(x, y)$, so $(1 - \alpha) d(x, y) \le 0$ and $x = y$. ■

Exercise 7.3 (§45, Ex. 3; Arzelà's theorem)

Let X be compact and $f_n \in \mathcal{C}(X, \mathbb{R}^k)$. Show that if $\{f_n\}$ is pointwise bounded and equicontinuous, then (f_n) has a uniformly convergent subsequence.

Solution.

Step 1. Let $\mathcal{F} = \{f_n\}$. By Ascoli's theorem (Theorem 7.12), the closure $\overline{\mathcal{F}}$ in $\mathcal{C}(X, \mathbb{R}^k)$ with the uniform topology is compact.

Step 2. The uniform topology comes from the uniform metric, so $\overline{\mathcal{F}}$ is a compact metrizable space, hence sequentially compact (Theorem 3.33).

Step 3. Therefore the sequence (f_n) in $\overline{\mathcal{F}}$ has a subsequence converging in the uniform metric, i.e. uniformly on X (for the bounded metric $\bar{d}$, uniform convergence with respect to $\bar{d}$ and to d are the same, since $\bar{d} = \min(d, 1)$). ■